

Probability and Statistics

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1 Introduction

Probability and Statistics are different things! Probability is the likelihood of an outcome. Statistics is the process of turning data into usable information. With this in mind let's come back to our favorite topic to ignore. Set Theory!

Now for this lecture notes, I will only require the understating provided in the introduction and notation document of set theory. That being, what a set is and what a subset is. For more depth on probability and statistics, you will need an understating of unions and intersections which are also covered, albeit not comprehensively, in the introduction.

But for us let's have the idea that a set is a collection of mathematical objects. This could be events, numbers, playing cards, or even ideas. A subset is a collection of mathematical objects that all already belong to a bigger set. (Note, subsets are sets themselves!) An example being if we take all real numbers $\mathbb{R}$ to be our set, then a subset would be the integers $\mathbb{Z}$.

2 Definitions and Theorems

Since the probability of an event is the likelihood of that event happening, we will define probability as a function that obeys certain axioms.

Probability

Let S be the set of all possible outcomes and $E \in S$. Then the probability of E is given by the function $P : S \rightarrow \mathbb{R}$ and obeys the following three axioms:

1. $0 \leq P(E) \leq 1$
2. $P(S) = 1$
3. For any sequence of mutually exclusive events we have the probability of the union of those events is equal to the sum of the probabilities of those events. Written:

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Quick note, while the third axiom is important to have a well defined definition we will not be using this property in the rest of this lecture notes.

Mean

The mean of a data set (or numerical set) is the average value that set takes denoted μ . If S is a set of numbers and $(a_i)_{i=1}^N$ is a sequence then we can write $S = \{a_i | i \in \mathbb{N}, a_i \in (a_i)_{i=1}^N\}$ Then the mean is written:

$$\mu = \frac{1}{N} \sum_{i=1}^N a_i$$

Median

Let $S = \{a_i | i \in \mathbb{N}, a_i \in (a_i)_{i=1}^N\}$ be a numerical set. Then the median is the middle value of the set. That is to say if we write S such that $S = \{a_1, a_2, \dots, a_N\}$ and each $a_1 \leq a_{i+1}$. Then the median M is

$$M = a_{\frac{N}{2}} \text{ if } N \text{ is even } \wedge \text{ the mean of the two middle values if } N \text{ is odd.}$$

Mode

Let $S = \{a_i | i \in \mathbb{N}, a_i \in (a_i)_{i=1}^N\}$ be a numerical set. Then the mode is the number occurring most often in the set.

A quick example of mode being: $S = \{1, 2, 3, 4, 3, 5, 2, 6, 3\}$ would have mode 3 as 3 occurs the most.

Distributions

A probability distribution is the probability measure.

This is an abstract definition and can be thought of like this. The probability distribution is the collection of all outcomes in relation to how likely an outcome is. This means that it can be a graph, a curve, an integral, and much more. The distribution simply must represent the data set.

(The word measure is a math word that requires significant mathematical understanding to get to. But can be thought of similar to what you would think when discussing an increasing one-to-one function)

3 Examples

Example 1: Find the mean, median, and mode for the given set S .

$$S = \{19, 23, 2, 11, 12, 19\}$$

$$M = \frac{1}{N} \sum_{i=1}^N a_i$$

Notice that S has 6 elements, so N is equal to 6.

$$\begin{aligned}\mu &= \frac{1}{N} \sum_{i=1}^N a_i \\ &= \frac{19 + 23 + 2 + 11 + 12 + 19}{6} \\ &= \frac{86}{6} \\ \mu &= 14.3\end{aligned}$$

For finding the median and mode, it is useful to organize the data set in ascending order.

$$S = \{2, 11, 12, 19, 19, 23\}$$

There are 6 data points, an even number, in the data set. This means we need to average the two data points closest

to the middle of the data set to find the median.

$$\begin{aligned}M &= \frac{12 + 19}{2} \\&= \frac{31}{2} \\M &= 15.5\end{aligned}$$

The mode is the value that appears most frequently in the data set. The only repeated data point in this set is 19. There are two 19s in the data set and only one of each other data point, so 19 is the mode.

Example 2: What is the probability of getting heads on a coin flip?

When flipping a coin, there are 2 possible outcomes for how it lands: heads or tails. With a fair coin, there are equal chances of flipping heads and tails. We now have enough information to solve the problem using a basic system of equations:

$$\begin{aligned}P(H) &= P(T) \\P(H) + P(T) &= 1\end{aligned}$$

Solving this simple system using substitution tells us that $P(H) = P(T) = 0.50$. Thus, there is a 50 percent probability of flipping heads.

Example 3: What is the probability of flipping heads, tails, heads?

For this problem, we are looking to flip a specific sequence: heads, tails, heads. To find the probability of a specific sequence, you multiply the probability of each individual step together. Let's consider each coin flip separately, remembering that $P(H) = P(T) = 0.50$.

First Flip: $P(H) = 0.50$

Second Flip: $P(T) = 0.50$

Third Flip: $P(H) = 0.50$

We will multiply the

$$\begin{aligned}
 P(HTH) &= P(H) * P(T) * P(H) \\
 &= 0.50 * 0.50 * 0.50 \\
 &= 0.125
 \end{aligned}$$

There is a 12.5 percent (1 in 8) probability of flipping heads, tails, heads.

Example 4: Two cards are chosen at random from a standard deck of 52 playing cards. What is the probability that they (a) have the same value? (b) are both aces? *Note this is a more advanced example that uses the 3rd axiom*

(a)

Consider choosing the cards one after another:

The first card can be absolutely any card in the deck. So given that there are 13 values in a standard 52 card deck, we will assume we chose a given card find the probability of that specific value. Then multiply by the 13 values.

- Then to start lets assume we have chosen a specific value. Then to start there are 4 of this value in the deck so the probability of picking one is $\frac{4}{52}$
- Once the first card is chosen, there are 51 cards remaining in the deck. Only 3 of those remaining cards match the value of the first card.

Thus, the probability that the second card matches the value of the first card is:

$$P(\text{same value}) = 13 \left(\frac{4}{52} \right) \left(\frac{3}{51} \right) = \frac{1}{17}$$

as we know we are going to have 13 possible first choices.

(b)

We can think of drawing the cards one after the other without replacement.

- Probability that the first card is an ace: $\frac{4}{52} = \frac{1}{13}$
- Probability that the second card is an ace (given the first was an ace): $\frac{3}{51} = \frac{1}{17}$

Multiplying these dependent probabilities yields:

$$P(\text{both aces}) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{13} \times \frac{1}{17} = \frac{1}{221}$$